

Homework 3: Latent Factors, Embeddings, and Attention

Math + Coding

Alexander Quispe Rojas
Universidad del Pacífico – MECA

Instructions. This homework covers Lectures 6 (latent factor models), 7 (embeddings), and 8 (attention). You may discuss the problems with classmates, but write up your own solutions. Show all work. Total: **120 points**.

Structure. (1) SVD & matrix factorization, (2) embeddings & word2vec, (3) attention & self-attention, (4) coding project (panel-data MF + self-attention from scratch).

Coding submission. Run `python _build_notebook.py` in `code/` to obtain `homework_3_student.ipynb`. Fill in the TODO cells. Submit the completed notebook alongside this PDF.

1 SVD and Matrix Factorization [30 Points]

Materials: Lecture 6 on latent factor models, SVD, PCA, matrix completion.

Problem 1 [4 Points]

Trace cyclic identity.

- (a) Prove $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ for $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times m}$.
- (b) Generalize: prove $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BCA}) = \text{tr}(\mathbf{CAB})$ for conformable matrices.

Problem 2 [4 Points]

SVD $\leftrightarrow$ PCA eigenvalues. Let $\mathbf{X} \in \mathbb{R}^{D \times N}$ have SVD $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ where $\mathbf{U} \in \mathbb{R}^{D \times D}$, $\mathbf{V} \in \mathbb{R}^{N \times N}$ are orthogonal and $\mathbf{\Sigma}$ is diagonal with singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

- (a) Show that $\mathbf{XX}^\top = \mathbf{U}\mathbf{\Sigma}^2\mathbf{U}^\top$. Conclude that the eigenvalues of $\mathbf{XX}^\top$ are exactly σ_d^2 .
- (b) Use this to argue that the eigenvalues of $\mathbf{XX}^\top$ are non-negative (no SVD required for the geometric reason).

Problem 3 [3 Points]

Truncated SVD storage. Let $\mathbf{X} \in \mathbb{R}^{N \times N}$ have rank k . Suppose we store its rank- k SVD ($\mathbf{U}_k \in \mathbb{R}^{N \times k}$, $\mathbf{\Sigma}_k$, $\mathbf{V}_k^\top$) instead of the full matrix.

- (a) How many real numbers do we need? (Hint: $\mathbf{\Sigma}_k$ is diagonal.)

Homework 3: Latent Factors, Embeddings, and Attention

- (b) For what values of k is the truncated SVD strictly smaller than storing $\mathbf{X}$ as a dense $N \times N$ matrix?

Problem 4 [5 Points]

Matrix factorization gradients. Consider the regularized objective

$$L(\mathbf{U}, \mathbf{V}) = \frac{\lambda}{2} (\|\mathbf{U}\|_F^2 + \|\mathbf{V}\|_F^2) + \frac{1}{2} \sum_{(i,j) \in S} (y_{ij} - \mathbf{u}_i^\top \mathbf{v}_j)^2$$

where $\mathbf{U} \in \mathbb{R}^{M \times K}$, $\mathbf{V} \in \mathbb{R}^{N \times K}$, $\mathbf{u}_i^\top$ is the i -th row of $\mathbf{U}$, and S is the set of observed entries.

Derive $\partial L / \partial \mathbf{u}_i$ and $\partial L / \partial \mathbf{v}_j$ (treat these as column vectors in $\mathbb{R}^K$).

Problem 5 [5 Points]

ALS closed form. Fix $\mathbf{V}$ and solve for the $\mathbf{u}_i$ that minimizes L . Derive the closed-form update.

Problem 6 [9 Points]

Panel data with one missing potential outcome.

Econ application of matrix completion (Athey, Bai, Imbens, Khosravi, 2021).

Suppose we have a 4×3 outcome panel $\mathbf{Y}$ for 4 firms (rows) over 3 time periods (columns). Firm 3 in period 2 received treatment, so $Y_{3,2}$ is the *observed treated* outcome; we want the counterfactual $Y_{3,2}(0)$ to compute the treatment effect.

The data:

$$\mathbf{Y} = \begin{bmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & ? & 9 \\ 4 & 8 & 12 \end{bmatrix} \quad (? = \text{missing/treated}).$$

Assume rank $K = 1$ and use latent factor model $Y_{ij} \approx u_i \cdot v_j$ where $u_i, v_j \in \mathbb{R}$. **Suppose the column factors are fixed** at $\mathbf{v} = (1, 2, 3)$ (e.g., from a long pre-treatment period).

- Using the ALS update from Problem 5 with $\lambda = 0$, compute the optimal u_i for each row i given only the observed entries.
- Use $\hat{u}_3$ and the fixed $\mathbf{v}$ to predict the counterfactual $\hat{Y}_{3,2}(0)$.
- The observed treated outcome is $Y_{3,2}(1) = 8$. Report the estimated treatment effect $\hat{\tau}_{3,2} = Y_{3,2}(1) - \hat{Y}_{3,2}(0)$.
- Why does this approach generalize the synthetic control method? Answer in 2-3 sentences.

2 Embeddings and Word2vec [30 Points]

Materials: Lecture 7 on embeddings, LLE, word2vec, hierarchical softmax, Huffman trees.

Problem 7 [5 Points]**Softmax gradient.**

The skip-gram conditional probability is

$$P(w_O | w_I) = \frac{\exp(\mathbf{v}'_{w_O} \mathbf{v}_{w_I})}{\sum_{w=1}^W \exp(\mathbf{v}'_w \mathbf{v}_{w_I})}$$

where $\mathbf{v}_w \in \mathbb{R}^D$ is the input embedding of word w and $\mathbf{v}'_w \in \mathbb{R}^D$ is its output embedding.

Derive $\nabla_{\mathbf{v}_{w_I}} \log P(w_O | w_I)$. Express the answer in terms of $\mathbf{v}'_{w_O}$, the model probabilities $P(w | w_I)$, and the output embeddings $\{\mathbf{v}'_w\}$. Give a one-sentence interpretation.

Problem 8 [3 Points]**Time complexity of full softmax.**

For one (w_I, w_O) pair, what is the time complexity of computing $\nabla_{\mathbf{v}_{w_I}} \log P(w_O | w_I)$ from scratch? Give the answer in terms of vocabulary size W and embedding dimension D , treating arithmetic operations as $O(1)$.

Problem 9 [9 Points]**Huffman tree vs balanced binary tree.**

Consider these 8 words with their frequencies from a small corpus:

Word	Count
<i>do</i>	18
<i>you</i>	4
<i>know</i>	7
<i>the</i>	20
<i>way</i>	9
<i>of</i>	4
<i>devil</i>	5
<i>queen</i>	6

Total: 73 occurrences.

- Build a Huffman tree for these words. Show your merging steps and report the resulting path length of each word. (Many valid trees; just produce *one*.)
- Build a balanced binary tree of depth 3 with all 8 words as leaves. What is the path length of each word?
- Compute the *expected* path length $\mathbb{E}[L] = \sum_w f_w L_w / \sum_w f_w$ under each tree. Which is shorter?
- In one sentence, explain why Huffman beats balanced for Zipfian (heavy-tailed) word distributions.

Problem 10 [5 Points]

Negative sampling gradient.

In negative sampling we replace the full softmax with a binary classification objective. For one observed pair (w_I, w_O) and k noise samples $\tilde{w}_1, \dots, \tilde{w}_k$, the loss is

$$\mathcal{L} = -\log \sigma(\mathbf{v}'_{w_O}{}^\top \mathbf{v}_{w_I}) - \sum_{i=1}^k \log \sigma(-\mathbf{v}'_{\tilde{w}_i}{}^\top \mathbf{v}_{w_I})$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid.

- (a) Derive $\nabla_{\mathbf{v}_{w_I}} \mathcal{L}$.
- (b) Compare your answer to the full-softmax gradient (Problem 7). What replaces the sum over all W output vectors?
- (c) State the per-pair time complexity in terms of k and D .

Problem 11 [8 Points]**LLE numerical example.**

Three collinear points in $\mathbb{R}^2$:

$$\mathbf{x}_1 = (0, 0), \quad \mathbf{x}_2 = (1, 1), \quad \mathbf{x}_3 = (2, 2).$$

Use $B = 2$ nearest neighbors throughout. Target dimension $K = 1$.

- (a) **Step 1.** For the interior point $\mathbf{x}_2$ with $\mathcal{B}(2) = \{1, 3\}$, find the weights $W_{2,1}, W_{2,3}$ subject to $W_{2,1} + W_{2,3} = 1$. Show your work.
- (b) **Step 2.** With $\mathbf{W}$ fixed, find low-dim coordinates $u_1, u_2, u_3 \in \mathbb{R}$ that satisfy the same convex-combination relation $u_2 = W_{2,1}u_1 + W_{2,3}u_3$, subject to (i) centering $\sum_i u_i = 0$ and (ii) unit variance $\frac{1}{3} \sum_i u_i^2 = 1$.
- (c) Verify your answer recovers the 1D ordering and that the local reconstruction holds.

3 Attention and Self-Attention [25 Points]

*Materials: Lecture 8 on attention, scaled dot-product, multi-head attention, masking.***Problem 12 [4 Points]****Scaled dot-product attention: shapes.**

Self-attention is given by

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}$$

where $\mathbf{Q} = \mathbf{x}\mathbf{W}_Q^\top$, $\mathbf{K} = \mathbf{x}\mathbf{W}_K^\top$, $\mathbf{V} = \mathbf{x}\mathbf{W}_V^\top$.The input $\mathbf{x} \in \mathbb{R}^{T \times D}$ has T tokens of embedding dim D . Key dim is d_k , value dim is d_v .

- (a) Give the shapes of $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$.
- (b) Give the shapes of $\mathbf{Q}, \mathbf{K}, \mathbf{V}$, and the score matrix $\mathbf{Q}\mathbf{K}^\top$.
- (c) Give the shape of the final output.

Problem 13 [8 Points]**Why $\sqrt{d_k}$? A CLT argument.**Assume the entries of $\mathbf{q}, \mathbf{k} \in \mathbb{R}^{d_k}$ are i.i.d. $\mathcal{N}(0, 1)$ and that $\mathbf{q}$ and $\mathbf{k}$ are independent. Define the dot-product score $s = \mathbf{q}^\top \mathbf{k}$.

- (a) Compute $\mathbb{E}[s]$.
- (b) Compute $\text{Var}(s)$.
- (c) Argue (via the CLT) that for large d_k , the distribution of s is approximately Gaussian. State its parameters.
- (d) Why does dividing s by $\sqrt{d_k}$ “fix” the softmax-saturation problem? Be specific about what variance you obtain after scaling.

Problem 14 [4 Points]**Multi-head attention with output projection.**In multi-head attention, h heads each compute

$$\mathbf{O}^i = \text{softmax}\left(\frac{\mathbf{Q}^i(\mathbf{K}^i)^\top}{\sqrt{d_k}}\right) \mathbf{V}^i$$

where $\mathbf{Q}^i = \mathbf{x}\mathbf{W}_Q^{i,\top}$, etc. The head outputs are concatenated and projected:

$$\mathbf{O} = [\mathbf{O}^1 \mathbf{O}^2 \dots \mathbf{O}^h] \mathbf{W}_O^\top.$$

The final output dimension is D .Give the shapes of $\mathbf{W}_Q^i, \mathbf{W}_K^i, \mathbf{W}_V^i, \mathbf{W}_O$ in terms of D, d_k, d_v, h .**Problem 15 [5 Points]**

Numerical attention example.

Take $T = 2$, $d_k = d_v = 2$. Given:

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}.$$

Compute the attention output $\mathbf{O} = \text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k})\mathbf{V}$ by hand, showing each step.

Problem 16 [4 Points]**Causal mask: explicit example.**

In a decoder language model, position i must not attend to positions $j > i$. The causal-masked score is

$$e_{i,j} = \begin{cases} \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d_k}, & j \leq i, \\ -\infty, & j > i. \end{cases}$$

Suppose $T = 4$ and the unmasked scaled scores are

$$\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}.$$

- (a) Write the score matrix after applying the causal mask.
- (b) For row $i = 2$ (zero-indexed: $i = 1$), write the resulting softmax row.

4 Coding Project [35 Points]

Materials: Lectures 6, 7, and 8. See `code/` for the notebook generator and `code/data/` for the `dr_seuss.txt` corpus.

Problem 17 [10 Points]**Matrix factorization on synthetic panel data.**

Open `homework_3_student.ipynb`. The notebook generates a synthetic $M = 20$ firms $\times$ $T = 12$ months panel of (log) sales with a known low-rank structure plus noise, then masks 25% of the entries to simulate a staggered-adoption treatment.

You will:

- Implement matrix factorization via SGD using your gradients from Problem 4. Initialize $\mathbf{U}, \mathbf{V}$ uniformly in $[-0.5, 0.5]$. Use learning rate $\eta = 0.03$.
- Sweep $K \in \{1, 2, 3, 5, 8, 12\}$ and plot in-sample loss E_{in} and held-out (masked) loss E_{out} vs K .
- Identify the optimal K . How close is it to the true rank of the data-generating process?
- Repeat with regularization $\lambda \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$. Briefly describe the bias-variance trade-off you observe.

Submit your completed notebook and a 4-line summary of part (d) here.

Problem 18 [5 Points]**Self-attention from scratch.**

In the same notebook, you will implement `SelfAttention` and `MultiHeadAttention` as PyTorch modules (no `nn.MultiheadAttention`).

- Implement single-head scaled dot-product self-attention. Verify your output matches `torch.nn.functional` to numerical tolerance ($< 10^{-5}$) on a random input.
- Extend to multi-head ($h = 4$) with an output projection $\mathbf{W}_O$. Confirm output shape is (T, D) matching the input.

Submit the completed cells in your notebook.

Problem 19 [10 Points]**Word2vec implementation on `dr_seuss.txt`.**

In the same notebook (Part 3), you will implement a working skip-gram model end-to-end and inspect the learned embeddings on Dr. Seuss text. The corpus is provided in `code/data/dr_seuss.txt` (2,071 tokens, ~ 308 unique words).

- Tokenize the corpus, build the vocabulary, and implement `generate_traindata(tokens, w2i, window_size=4)` that returns `(trainX, trainY)` as arrays of one-hot vectors – one training pair per (center, context) inside a sliding window of half-width $s = 4$.
- Build a 2-layer linear network $\text{Linear}(W \rightarrow D) \rightarrow \text{Linear}(D \rightarrow W)$ (no activation in the hidden layer – this is the textbook skip-gram architecture). Use cross-entropy loss and Adam (lr = 0.01) for 500 epochs. Set $D = 10$.

Homework 3: Latent Factors, Embeddings, and Attention

- (c) ~~Extract the hidden-layer weight matrix as the embedding (*shape conventions: `nn.Linear.weight`* is `(out, in)`; transpose to get one row per word). Compute pairwise cosine similarities and report the top 30 most-similar word pairs (deduplicated).~~
- (d) Briefly answer in your write-up: (i) what are the dimensions of the hidden-layer and output-layer weight matrices? (ii) Do the resulting pairs make sense? (iii) Why do so many of the top pairs look like *rhymes*?

Problem 20 [10 Points]**Applied Transformer: sentiment + attention visualization.**

Up to now we built attention modules from scratch. In Part 4 of the notebook you will work with a *real, pretrained Transformer*: `distilbert-base-uncased-finetuned-sst-2-english` (HuggingFace).

The model is 6 layers, 12 heads per layer, hidden dim 768, fine-tuned on the SST-2 binary-sentiment task. First run downloads ~250 MB; everything runs on CPU in seconds.

- Load the model and tokenizer. Run sentiment inference on five finance-flavored sentences (provided). Report the predicted label and confidence for each.
- Pick one of the five sentences. Extract the attention tensor from `outputs.attentions` – shape (layers, batch, heads, T , T). Plot the attention map for layer 3, head 0 as a heatmap, with token strings on both axes.
- In a markdown cell, answer: (i) was the model’s prediction consistent with a human judgement? (ii) Looking at the heatmap, are the attention patterns interpretable – do tokens attend to plausibly related tokens? (iii) Try a sentence with *ambiguous* sentiment (e.g., “*The hike was widely expected but the press conference was hawkish*”). Does the model still produce a confident label? What changes in the attention map?

End of Homework 3. *120 points total: Section 1 (30) + Section 2 (30) + Section 3 (25) + Section 4 (35).*